

Ch31: 3, 17, 23, 26, 30, 43, 58

Basic score 4` + problem 30 2.5`+ problem 43 2.5`+ problem 58 1`

3. (a) Since $L_{\text{eq}} = L_1 + L_2$ and $C_{\text{eq}} = C_1 + C_2 + C_3$ for the circuit, the resonant frequency is

$$f = \frac{1}{2\pi\sqrt{L_{\text{eq}}C_{\text{eq}}}} = \frac{1}{2\pi\sqrt{(L_1+L_2)(C_1+C_2+C_3)}}$$

$$= \frac{1}{2\pi\sqrt{(9.70 \times 10^{-3} \text{ H} + 2.30 \times 10^{-3} \text{ H})(8.40 \times 10^{-6} \text{ F} + 2.50 \times 10^{-6} \text{ F} + 3.50 \times 10^{-6} \text{ F})}}$$

$$= 383 \text{ Hz}$$

(b) The resonant frequency does not depend on R so it will not change as R increases.

(c) Since $\omega \propto (L_1 + L_2)^{-1/2}$, it will decrease as L_1 increases.

(d) Since $\omega \propto C_{\text{eq}}^{-1/2}$ and C_{eq} decreases as C_3 is removed, ω will increase.

(e) Since $\omega \propto (L_1 + L_2)^{-1/2}$, it will increase as L_2 is removed.

17. (a) The generator emf and the current are given by

$$\mathcal{E} = \mathcal{E}_m \sin(\omega_d t - \pi/4), \quad i(t) = I \sin(\omega_d t - 3\pi/4).$$

The expressions show that the emf is maximum when $\sin(\omega_d t - \pi/4) = 1$

The first time this occurs after $t = 0$ is when $\omega_d t - \pi/4 = \pi/2$ (that is, $n = 0$). Therefore,

$$t = \frac{3\pi}{4\omega_d} = \frac{3\pi}{4(270 \text{ rad/s})} = 8.73 \times 10^{-3} \text{ s}.$$

(b) The current is maximum when $\sin(\omega_d t - 3\pi/4) = 1$,

The first time this occurs after $t = 0$ is when $\omega_d t - 3\pi/4 = \pi/2$ (as in part (a), $n = 0$). Therefore,

$$t = \frac{5\pi}{4\omega_d} = \frac{5\pi}{4(270 \text{ rad/s})} = 1.45 \times 10^{-2} \text{ s}.$$

(c) The current lags the emf by $+\pi/2$ rad, so the circuit element must be an inductor.

(d) The current amplitude I is related to the voltage amplitude V_L by $V_L = IX_L$, where X_L is the inductive reactance, given by $X_L = \omega_d L$. Furthermore, since there is only one element in the circuit, the amplitude of the potential difference across the element must be the same as the amplitude of the generator emf: $V_L = \varepsilon_m$. Thus, $\varepsilon_m = I\omega_d L$ and

$$L = \frac{\varepsilon_m}{I\omega_d} = \frac{(25\text{V})}{(0.62\text{A})(270\text{rad/s})} = 0.149 \text{ H}$$

Note: The current in the circuit can be rewritten as

$$i(t) = I \sin\left(\omega_d t - \frac{3\pi}{4}\right) = I \sin\left(\omega_d t - \frac{\pi}{4} - \phi\right)$$

where $\phi = +\pi/2$. In a purely inductive circuit, the current lags the voltage by 90° .

23. (a) The power factor is $\cos \phi$, where ϕ is the phase constant defined by the expression $i = I \sin(\omega t - \phi)$. Thus, $\phi = -42.0^\circ$ and $\cos \phi = \cos(-42.0^\circ) = 0.743$.

(b) Since $\phi < 0$, $\omega t - \phi > \omega t$. The current leads the emf.

(c) The phase constant is related to the reactance difference by $\tan \phi = (X_L - X_C)/R$. We have

$$\tan \phi = \tan(-42.0^\circ) = -0.900,$$

a negative number. Therefore, $X_L - X_C$ is negative, which leads to $X_C > X_L$. The circuit in the box is predominantly capacitive.

(d) If the circuit were in resonance X_L would be the same as X_C , $\tan \phi$ would be zero, and ϕ would be zero. Since ϕ is not zero, we conclude the circuit is not in resonance.

(e) Since $\tan \phi$ is negative and finite, neither the capacitive reactance nor the resistance are zero. This means the box must contain a capacitor and a resistor.

(f) The inductive reactance may be zero, so there need not be an inductor.

(g) Yes, there is a resistor.

(h) The average power is

$$P_{\text{avg}} = \frac{1}{2} \varepsilon_m I \cos \phi = \frac{1}{2} (61.4\text{V})(0.93\text{A})(0.743) = 21.3\text{W}$$

(i) The answers above depend on the frequency only through the phase constant ϕ , which is given. If values were given for R , L and C then the value of the frequency would also be needed to compute the power factor.

26. (a) With both switches closed (which effectively removes the resistor from the circuit), the impedance is just equal to the (net) reactance and is equal to

$$X_{\text{net}} = X_L - X_C = (12 \text{ V})/(0.447 \text{ A}) = 26.85 \Omega.$$

With switch 1 closed but switch 2 open, we have the same (net) reactance as just discussed, but now the resistor is part of the circuit; using Eq. 31-65 we find.

$$\tan \phi = \frac{X_L - X_C}{R} = \frac{X_{\text{net}}}{R} \Rightarrow R = \frac{X_{\text{net}}}{\tan \phi} = \frac{26.85 \Omega}{\tan 20^\circ} = 73.77 \Omega$$

(b) For the first situation described in the problem (both switches open) we can reverse our reasoning of part (a) and find

$$X'_{\text{net}} = X_L - X'_C = R \tan \phi' = (73.77 \Omega) \tan(-25.0^\circ) = -34.40 \Omega.$$

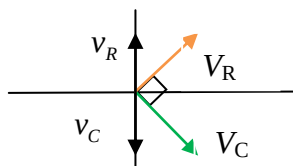
We observe that the effect of switch 1 implies

$$X_{\text{net}} - X'_{\text{net}} = X'_C - X_C = X_C = 26.85 \Omega - (-34.40 \Omega) = 61.25 \Omega.$$

Then Eq. 31-39 leads to $C = 1/2\pi f X_C = 43.3 \mu\text{F}$.

(c) Since $X_{\text{net}} = X_L - X_C$, then we find $L = X_L/2\pi f = 234 \text{ mH}$.

30. (a)



0.5point (V_R 和 V_C 相对相位关系正确即可)

(b) We have $IR = IX_C$, or

$$IR = IX_C \rightarrow R = \frac{1}{\omega_d C}$$

which yields $f = \frac{\omega_d}{2\pi} = \frac{1}{2\pi RC} = \frac{1}{2\pi(50.0 \Omega)(2.80 \times 10^{-5} \text{ F})} = 114 \text{ Hz}$. 0.5 point

(c) $\phi = \tan^{-1}(-V_C/V_R) = -45^\circ$. 0.5 point

(d) $\omega_d = 1/RC = 7.14 \times 10^2 \text{ rad/s}$. 0.5 point

(e) $I = (12 \text{ V})/\sqrt{R^2 + X_C^2} = 6/(25\sqrt{2}) \approx 170 \text{ mA}$. 0.5 point

以上均公式 0.3 point, 结果 0.2 point

43. (a) The total energy U is the sum of the energies in the inductor and capacitor:

$$U = U_E + U_B = \frac{q^2}{2C} + \frac{i^2 L}{2} = \frac{(3.80 \times 10^{-6} \text{ C})^2}{2(2.89 \times 10^{-6} \text{ F})} + \frac{(9.20 \times 10^{-3} \text{ A})^2 (25.0 \times 10^{-3} \text{ H})}{2} = 3.56 \times 10^{-6} \text{ J}$$

公式 0.3 point 结果 0.2 point

(b) We solve $U = Q^2/2C$ for the maximum charge:

$$Q = \sqrt{2CU} = \sqrt{2(2.89 \times 10^{-6} \text{ F})(3.56 \times 10^{-6} \text{ J})} = 4.54 \times 10^{-6} \text{ C}$$

公式 0.3 point 结果 0.2 point

(c) From $U = I^2 L/2$, we find the maximum current:

$$I = \sqrt{2U/L} = \sqrt{2(3.56 \times 10^{-6} \text{ J})/(25.0 \times 10^{-2} \text{ H})} = 1.69 \times 10^{-2} \text{ mA}$$

公式 0.3 point 结果 0.2 point

(d) If q_0 is the charge on the capacitor at time $t = 0$, then $q_0 = Q \cos \phi$ and

$$\phi = \cos^{-1}\left(\frac{3.80}{4.54}\right) = \pm 33.2^\circ \quad 0.5 \text{ point}$$

For $\phi = +33.2^\circ$ the charge on the capacitor is decreasing, for $\phi = -33.2^\circ$ it is increasing. To check this, we calculate the derivative of q with respect to time, evaluated for $t = 0$. We obtain $-\omega Q \sin \phi$, which we wish to be positive. Since $\sin(+33.2^\circ)$ is positive and $\sin(-33.2^\circ)$ is negative, the correct value for increasing charge is $\phi = -33.2^\circ$.

0.3 point

(e) Now we want the derivative to be negative and $\sin \phi$ to be positive. Thus, we take $\phi = +33.2^\circ$

0.2 point

58. (a) The graph shows that the resonance angular frequency is 25000 rad/s, which means (using Eq. 31-4)

$$C = (\omega^2 L)^{-1} = [(25000)^2 \times 450 \times 10^{-6}]^{-1} = 3.56 \mu\text{F}. \quad \text{公式 0.3 point 结果 0.2 point}$$

(b) The graph also shows that the current amplitude at resonance is $I_s = 4.0 \text{ A}$, but at resonance the impedance Z becomes purely resistive ($Z = R$) so that we have:

$$I_s = \frac{\varepsilon_m}{R} \Rightarrow R = \frac{\varepsilon_m}{I_s} = \frac{6.0 \text{ V}}{4.0 \text{ A}} = 1.5 \Omega$$

公式 0.3 point 结果 0.2 point